

BENARBIA EMIR

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Data Engineering & AI student building RAG systems, LLM agents, and data pipelines

Seeking 6-Month Final Year Internship (French: PFE) | Jan – Jul 2027 | Open to Relocation

Education

National Engineering Degree – Data & Software Engineering

Sep. 2023 – Jun. 2027

Faculty of Sciences of Bizerte · University of Carthage

Bizerte, Tunisia

- Ranked top 3 of 36 in the Data & Software Engineering program

Experience

Data & Software Engineering Intern

Jun. 2026 – Aug. 2026

TcodeX

Tunis, Tunisia

- Engineered MediDocs, a full-stack clinical document extraction platform (**Django**, **PostgreSQL**, **React/Vite**) automating structured JSON ingestion from medical scans.
- Benchmarked local and API-hosted **vision-language models**, including **Nemotron-3-Nano-Omni-30B** via **OpenRouter**, in **Kaggle** before deploying the best-performing **OCR** pipeline to production.
- Built an automated extraction-quality scoring system (**scikit-learn** TF-IDF/cosine similarity, difflib edit-distance) to quantify human-in-the-loop correction effort.
- Designed schema-agnostic **React** components for dynamic form rendering; implemented session-based auth with **CSRF** protection across a cross-origin architecture.

Software Engineering Intern

Jun. 2025 – Aug. 2025

Générale Tunisienne de l'Informatique (GTI)

Tunis, Tunisia

- Built a full-stack credit request management platform (**Angular**, **Spring Boot**, **PostgreSQL**) serving **3 user roles** across **2 client types**, cutting application processing time by **40%**.
- Designed **RESTful APIs** covering the end-to-end credit lifecycle (submission, validation, approval) with **JUnit** test coverage, replacing manual workflows.
- Built a responsive **Angular** frontend with dynamic forms, real-time status tracking, and **role-based access control (RBAC)**.
- Deployed a local conversational assistant (**Ollama**) to guide users through credit applications, reducing support-ticket volume.

Projects

Fuel Observatory & Monitoring System | Python, FastAPI, PostgreSQL, RAG, LLM Agents

April 2026

- Built a **real-time** fuel-market monitoring platform with multi-source ingestion and automated alerting.
- Added an **LLM agent** with **RAG** for conversational querying over structured data.

Global Education Intelligence Platform (BI & Data Warehouse) | Python, PostgreSQL, Django, Docker, GitHub Actions

- Designed a **star-schema data warehouse** fed by a modular, dependency-aware **ETL pipeline**.
- Exposed it via a **Django REST API** with **React/Streamlit** front ends, containerized with **Docker** + **CI/CD**.

Credit Card Fraud Detection | Python, Scikit-learn, Pandas, Matplotlib

December 2025

- Built a fraud-detection pipeline for **highly imbalanced** data using **supervised ML**.
- Applied **SMOTE**/undersampling, improving minority-class **F1-score** by **25%**.

Technical Skills

AI & Data: LLM Agents, RAG, ChromaDB, Groq, PyTorch, Scikit-learn, NLP, Pandas, NumPy

Data Engineering: ETL Pipelines, Data Warehousing (star schema), MLOps (MLflow, DVC), Streamlit

Languages: Python, SQL, Java, Go, JavaScript, C++

Backend & Web: Django, FastAPI, Spring Boot, React, Angular, REST APIs, Flask

Databases & DevOps: PostgreSQL, MySQL, MongoDB, Docker, CI/CD, Git, Linux, Postman

Extracurricular Activities

- Vice President & Social Media Manager** — Interact Club North Africa. Led 15+ members; grew engagement through social content strategy.
- Media Assistant** — TPL 2025 Tournament. Produced live coverage and real-time communications for a multi-day tournament.

Languages

Arabic: Native | **French:** Full Professional Proficiency (Fluent) — language of academic instruction | **English:** Full Professional Proficiency (Fluent) — technical documentation, coursework, and professional communication